

AgriSLM Ethical Dataset Data Card

South Africa + Sri Lanka Joint Agricultural Small Language Model (AgriSLM) TRI AI – Cohort 10 | Challenge 2 – Data Card

Final Draft – aligned to the Values-Led Problem Statement and Data Card Task

1. Values-Led Problem and Dataset Purpose

AgriSLM addresses agricultural information inequality affecting selected resource-constrained farming communities in South Africa and Sri Lanka. Farmers may have difficulty accessing timely, localized, scientifically grounded and understandable agricultural guidance, especially where connectivity, language, extension access and resources are constrained. The dataset is designed to support a compact, regionally grounded, multilingual and offline-capable agricultural language model that can provide safe agricultural decision support without replacing farmers, extension officers or agricultural experts.

The dataset is guided by equity, accessibility, scientific integrity, cultural preservation, sustainability, transparency and human agency. It prioritizes authoritative agricultural and climate knowledge while creating a controlled, consent-based pathway for incorporating farmer experience and validated local knowledge. Community members should benefit not merely as data sources, but as participants in defining needs, reviewing usefulness and identifying harms.

2. Input vs. Output

User Inputs

- Text questions in supported languages
- Descriptions of crop and farming challenges
- Crop, crop-stage and seasonal context
- Soil, water and land-management concerns
- Climate and weather-related concerns
- Questions about pests and diseases
- Questions about fertilizer and sustainable production
- Descriptions of relevant traditional/local practices
- Resource and connectivity constraints where relevant

Model Outputs

- Localized, context-specific agricultural guidance
- Source-grounded agronomic information
- Interpretation of trusted weather/climate information for agricultural decisions
- Risk-aware guidance for crop, soil, water, pest and disease challenges
- Safety-controlled responses for chemical or other high-risk queries
- Clear uncertainty or escalation messages when evidence is insufficient
- Source/provenance information where technically available
- Region, crop, language and risk metadata used for evaluation and governance

The AgriSLM is not intended to independently forecast weather. Trusted meteorological or agrometeorological data will be interpreted in agricultural context.

3. Classification Problem

AgriSLM is primarily an advisory-generation system, but classification supports retrieval, safety, localization and evaluation. Proposed classes include:

- Country and region
- Agro-ecological zone
- Language
- Crop and crop stage
- Knowledge domain
- Soil/terrain context where supported
- Season
- Climate event
- Pest or disease
- Practice type
- Evidence status
- Risk level and safety flag
- Regulatory context where relevant

Land tenure, labour type and resource access may be retained only when genuinely relevant to context or research analysis. They must not be used as proxies for farmer capability, worth, ethnicity or entitlement.

4. Labeling

Labels will come from a combination of authoritative documentation, domain experts and consented community contributions.

- Authoritative agricultural manuals and official crop guidance
- Official climate/agrometeorological information
- Agronomists and agricultural extension practitioners
- Peer-reviewed research where appropriate
- Community/farmer knowledge collected with consent and separately identified by evidence status

Consistency and accuracy checks will include source cross-checking, expert review for high-risk content, deduplication, metadata validation, provenance checks and manual review of safety-sensitive records.

Local knowledge will use three evidence labels: A – scientifically supported; B – locally observed/not yet scientifically validated; C – contradicted or potentially unsafe. Category B must never be silently presented as Category A.

5. Data Creation

The dataset plan separates data by function so that training/adaptation, retrieval and evaluation are not mixed.

Dataset component	Purpose
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Authoritative Knowledge Corpus	Official crop, soil, water, fertilizer, pest/disease and sustainability knowledge.
Climate/Agrometeorological Data	Trusted weather and climate information for contextual interpretation.
Validated Local Knowledge Corpus	Consented farmer/community knowledge with evidence-status and provenance labels.
Instruction/Adaptation Dataset	Curated question-answer or instruction examples used only if model adaptation is justified.
Held-Out Evaluation Dataset	Controlled questions excluded from training/adaptation and used to compare model configurations.

Primary Data – Planned

- Structured or semi-structured farmer interviews
- Focus-group discussions where appropriate
- Extension-practitioner and agricultural-expert consultation
- Field observations where ethically approved and useful
- Explicit informed consent before collection
- Clear explanation of purpose, use, storage, withdrawal options and limitations

No claim is made that these interviews or focus groups have already been completed; they are Phase 1 data-collection activities.

Secondary Data – Planned/Authoritative Sources

- South African national/provincial agricultural authorities
- Agricultural Research Council (South Africa)
- Official meteorological/agrometeorological sources
- Sri Lanka Department of Agriculture and relevant research divisions
- Sri Lanka meteorological/agrometeorological authorities
- Universities and peer-reviewed literature
- Other validated international sources where regional sources are insufficient

6. Web Collection Ethics

- Use official, authorized and publicly accessible sources wherever possible.
- Record source organization, title, date/version, region, language, usage rights and retrieval date.
- Do not assume that public accessibility automatically grants unrestricted training rights.
- Review copyright, licensing and terms before copying material into training/adaptation corpora.
- Do not collect private or personally identifying farmer information without a defined need and consent.
- Do not scrape closed communities, private groups or culturally sensitive knowledge without permission.
- AI tools may assist cleaning, formatting or quality checks, but synthetic farmer testimony must not be represented as authentic community data.
- Maintain a source register so contested or outdated records can be corrected or removed.

7. Bias and Representation

Potential Exclusions

- Farmers outside selected pilot regions
- Farmers growing crops outside the Phase 1 crop scope
- Speakers of languages not selected for Phase 1
- Women, youth, elderly farmers, landless/tenant farmers or other groups if recruitment is not inclusive
- Farmers with limited digital access
- Traditional/local knowledge that is poorly documented
- Less common pest, disease or climate events
- Communities whose agricultural conditions differ from the dominant records in the dataset

Mitigation

- Document geographic, crop and language boundaries clearly.
- Recruit diverse farmer and extension voices during field validation.
- Track dataset composition by country, region, language, crop, knowledge domain and risk level.
- Use metadata to prevent national or regional overgeneralization.
- Evaluate performance separately across represented groups rather than relying only on one overall score.
- Expand languages, crops and regions only after Phase 1 evaluation identifies priorities.
- Publish known limitations and representation gaps with each dataset version.

8. Values and Community Alignment

Value	Dataset Design Commitment
Equity	Prioritize information needs of resource-constrained farmers and measure representation rather than assuming equal coverage.
Accessibility	Support appropriate local languages, understandable outputs and low-connectivity/offline deployment.
Scientific integrity	Prioritize authoritative evidence, provenance, expert review and explicit uncertainty.
Cultural preservation	Include consented local knowledge without misrepresenting unvalidated practice as scientific fact.
Sustainability	Prioritize soil, water, climate resilience and responsible production knowledge.
Transparency	Document origins, licenses, transformations, limitations, exclusions and dataset versions.
Human agency	Support farmer and extension decision-making rather than replacing professional or community judgment.

Community benefit will be assessed through usefulness, understandability, accessibility and safety—not simply model performance. Accountability will include farmer and extension feedback, expert review, documented limitations, correction/removal procedures, dataset versioning and clear escalation for high-risk advice.

9. Producers, Agents and Users

Source producers: Official institutions, published researchers and consented farmers/community contributors whose knowledge is represented.

Core producers: QueenElizabeth/AgriSLM research team responsible for dataset planning, documentation, curation and versioning.

Adjacent producers: Agronomists, extension practitioners, translators/language reviewers and other approved advisors supporting collection, labeling or interpretation.

Core agents: Researchers and developers using the dataset to build, retrieve, evaluate or improve AgriSLM.

Impacted agents: Agricultural experts, extension services and partner institutions whose work may be affected by deployment or dataset updates.

Typical users: Selected farmers and agricultural support users who receive AgriSLM decision-support information.

Impacted users: Farmers or communities who could experience meaningful benefit or harm from inaccurate, exclusionary or inappropriate guidance.

Contributors: Consenting farmers, extension practitioners or experts who deliberately contribute data or evaluation feedback.

10. Metadata and Provenance Schema

Recommended fields:

record_id; country; region; agro_ecological_zone; language; crop; crop_stage; knowledge_domain; soil_type; terrain; season; climate_event; pest; disease; practice_type; risk_level; safety_flag; source_type; source_authority; source_reference; source_date; usage_rights; evidence_status; local_knowledge_status; consent_status; validation_status; reviewer_role; regulatory_context; last_reviewed.

11. Safety and Risk Handling

Risk level	Example	Dataset/model requirement
Low	General planting-period information	Normal grounded response.
Medium	Possible causes of crop symptoms	Provide alternatives and uncertainty.
High	Pesticide/fertilizer rate or other potentially harmful instruction	Require authoritative source grounding and relevant regulatory context.
Critical / unsupported	Dangerous chemical use or uncertain high-consequence diagnosis	Do not provide unsupported instruction; escalate to appropriate expert/authority.

12. Dataset Splitting and Evaluation

The prototype should maintain a held-out evaluation set of approximately 200–500 controlled questions, balanced where feasible across country, region, crop, language, knowledge domain and risk level. Evaluation records must not be used in model adaptation or training.

- Baseline A: unadapted compact/base SLM
- Baseline B: same model with approved agricultural RAG
- Experimental C: adapted AgriSLM plus RAG, if adaptation is justified

Evaluation should measure agronomic correctness, groundedness/source attribution, regional relevance, hallucination/unsupported claims, safety, language accessibility, usefulness and offline technical performance.

13. Intended Uses

- Research and evaluation of localized agricultural language-model decision support
- Retrieval-grounded answers for selected crops and regions
- Climate-aware interpretation using trusted external climate information
- Support for farmer and extension education and information access
- Evaluation of multilingual and offline-capable agricultural AI

14. Out-of-Scope / Prohibited Uses

- Automated land allocation, land-reform or entitlement decisions
- Replacing agricultural extension officers, agronomists or relevant authorities
- Guaranteeing crop yield, income or climate outcomes
- Unsupported pesticide, chemical, veterinary, legal or financial instructions
- Profiling farmer capability or eligibility from ethnicity, land tenure, language or socioeconomic status
- Using community knowledge beyond the consent, license or governance conditions under which it was contributed

15. Accountability, Maintenance and Versioning

- Maintain dataset version numbers and change logs.
- Record source additions, corrections and removals.
- Periodically review time-sensitive agronomic, regulatory and climate information.
- Provide a process for community members or experts to flag inaccurate, harmful or misrepresented records.
- Re-evaluate bias and representation after major dataset expansion.
- Document known limitations with each release.
- Use external agronomic review for safety-critical areas where practical.

16. Phase 1 Scope and Open Research Variables

Current research direction:

- Sri Lanka pilot: Nuwara Eliya; priority highland crops proposed to include potato, beans and tomato/selected vegetables; Tamil and English proposed for Phase 1.
- South Africa pilot: one selected Gauteng agricultural community/district is proposed; exact municipality, priority crops and local African language remain Phase 1 validation decisions.
- Knowledge domains: crop production, soil and water management, climate interpretation, pests/diseases, fertilizer guidance and sustainable production.
- Model direction: compact open-weight SLM, retrieval-first architecture, with fine-tuning/adaptation only if benchmarking demonstrates benefit.

These unresolved items are research variables and should not be presented as completed field findings.

17. Known Limitations

- The dataset will initially represent only selected communities, crops and languages.
- Farmer requirements and language preferences require field validation.
- Authoritative sources may differ in update frequency, terminology and regional specificity.
- Traditional/local knowledge may be difficult to validate or document without losing context.
- Low-resource languages may have limited high-quality agricultural text.
- Offline deployment imposes model-size and update constraints.
- Dataset quality cannot remove wider structural constraints such as land, finance, infrastructure, markets or extension capacity.

18. Community Benefit and Accountability Statement

The dataset is justified only if it contributes to genuine community benefit. AgriSLM therefore treats farmers as stakeholders in the research rather than merely as data producers. The project will seek to make agricultural knowledge more accessible while preserving the right of communities to understand how their knowledge is used, to challenge inaccurate representation, and to participate in evaluating whether the system is useful and safe. Success will be judged by grounded agricultural correctness, accessibility, regional relevance, safety and demonstrated usefulness to intended communities.

19. Template Attribution

This Data Card is structured in response to the TRI AI task 'Build a dataset ethically with a Data Card' and retains the responsible-innovation direction of the team's earlier Data Card. The approach is informed by the Data Cards framework referenced in the task material. This document is a research dataset plan and should be revised as Phase 1 evidence, permissions, community feedback and technical evaluation become available.